

1. Administrative & Study Identification

Full Study Title: A Phase 3, Multicenter, Randomized, Double-Blind, Placebo-Controlled Study to Evaluate the Efficacy and Safety of Enlicitide Decanoate Coadministered With Rosuvastatin in Adults With Hyperlipidemia **Short Title/Acronym:** A Clinical Study of Enlicitide in Participants With High Cholesterol (MK-0616-037) **Protocol Number:** MK-0616-037 **NCT Number:** NCT07216482 **Sponsor Name:** Merck (Merck Sharp & Dohme LLC) **Investigational Product:** Enlicitide Decanoate (coadministered with Rosuvastatin) **Phase of Development:** Phase 3 **Medical Monitor:** Merck Sharp & Dohme Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate if enlicitide coadministered with rosuvastatin works better than placebo on lowering low-density lipoprotein cholesterol (LDL-C) in a participant's blood at Week 8. **Secondary Objectives:** To evaluate the mean percent change from baseline in LDL-C and apolipoprotein B (ApoB) at Week 8, and to assess the overall safety and tolerability of the combination therapy by monitoring adverse events (AEs).

2.2 Background & Rationale Hypercholesterolemia, a type of hyperlipidemia, affects approximately 86 million adults in the U.S. and is a major risk factor for atherosclerotic cardiovascular disease (ASCVD). Nearly 70% of people with ASCVD treated with standard lipid-lowering therapies do not reach target LDL-C levels. Enlicitide is an investigational oral PCSK9 inhibitor designed to lower LDL-C via the same biological mechanism as injectable PCSK9 inhibitors but in a convenient daily pill form. This study aims to determine the efficacy and safety of coadministering enlicitide with rosuvastatin to help close gaps in lipid management.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled **Blinding:** Double-blind **Randomization:** Randomized **Trial Status:** Not Yet Recruiting

3.2 Planned Enrollment & Duration Target N\$: 975 adult participants **Number of Sites:** Multicenter (Approx. 88 locations globally) **Estimated Study Start:** 11-Nov-2025 **Estimated Primary Completion:** 02-Dec-2026

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age 18 to 64 years. 2. Diagnosed with Hyperlipidemia. 3. Has either not received lipid-lowering therapy (LLT) or has not received certain LLTs within a specified time period prior to the study.

4.2 Exclusion Criteria 1. Has a history of homozygous familial hypercholesterolemia (FH), compound heterozygous FH, or double heterozygous FH. 2. Had a heart failure hospitalization within 3 months before Screening. 3. Is unwilling to take a statin, and/or has a history of statin-associated muscle symptoms or other statin-related AEs to any statin and dose, and/or is known to be intolerant to 1 or more statins. 4. Has a history of any of the following conditions: (1) Myopathy, myositis, rhabdomyolysis, or unexplained muscle pain; (2) Muscular or neuromuscular disease; (3) Neuropathy, fibromyalgia, or chronic pain; or (4) Has a personal or family history of hereditary muscular disorders. 5. Has active or chronic hepatobiliary or hepatic disease. 6. Has known human immunodeficiency virus (HIV) infection. 7. Is undergoing or previously underwent an LDL-C apheresis program within 3 months before Screening or plans to initiate an LDL-C apheresis program. 8. Has received any medication that may interact with rosuvastatin within 5 half-lives prior to Screening or is planning to initiate such treatment.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Once-daily oral pill **Route of Administration:** Oral **Duration of Treatment:** Base treatment up to 24-52 weeks (Primary endpoint measured at 8 weeks). **Concomitant Background Medication:** Rosuvastatin (Trade names: Crestor, Ezallor, Roszet | ATC Code: C10AA07).

5.2 Concomitant & Prohibited Therapy Permitted: Rosuvastatin as a standard background lipid-lowering therapy. **Prohibited:** Other lipid-lowering therapies (e.g., injectable PCSK9 inhibitors) not specified in the protocol; medications that interact with rosuvastatin within 5 half-lives prior to Screening.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -14 to 0	Informed Consent, Medical History, Fasting Lipid Panel (LDL-C)
Baseline	Day 0	Randomization, First Dose, Dispense Study Medication
Interim 1	Week 8	Primary Efficacy Assessment (Blood sample for LDL-C), Safety Labs
End of Study	Week 24 / 52	Final Vital Signs, Exit Interview, IP Accountability, Final Lipid Assessment

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Mean Percent Change from Baseline in LDL-C at Week 8 (Enlicitide + Rosuvastatin Versus Placebo). **Measurement Method:** Blood samples will be collected at baseline and at Week 8 to assess mean percent change in LDL-C between participants treated with enlicitide and rosuvastatin versus placebo.

7.2 Secondary Endpoints 1. Mean Percent Change from Baseline in LDL-C at Week 8. 2. Mean Percent Change from Baseline in Apolipoprotein B (ApoB) at Week 8. 3. Number of Participants with One or More Adverse Events (AEs).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Patient diary, HCP interviews at each visit; monitoring for statin-associated muscle symptoms and hepatic events. **Serious Adverse Events (SAEs):** Standard reporting timeline, typically within 24 hours of site awareness. **Data Monitoring Committee (DMC):** Independent external DMC to oversee safety and efficacy data throughout the trial.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for primary efficacy analysis; Safety Set for all participants who receive at least one dose of study medication. **Sample Size Justification:** Target $N=975$ appropriately powered to detect a statistically significant superiority in LDL-C reduction of enlicitide + rosuvastatin over placebo + rosuvastatin. **Handling of Missing Data:** Multiple Imputation (MI) and model-based approaches for continuous lipid variables.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Routine onsite and remote monitoring visits to ensure protocol compliance and data integrity. **Audit Trail:** Secure eCRF systems with audit trails, data clarification forms, and automated edit checks.

11. Study Identifiers & Governance

Primary ID: MK-0616-037 **Posting ID:** 367580228062351295 **Registry Number:** NCT07216482 **Sponsor/Lead Organization:** Merck **Study Title:**

A Clinical Study of Enlicitide in Participants With High Cholesterol (MK-0616-037) **Official Regulatory Title:** A Phase 3, Multicenter, Randomized, Double-Blind, Placebo-Controlled Study to Evaluate the Efficacy and Safety of Enlicitide Decanoate Coadministered With Rosuvastatin in Adults With Hyperlipidemia

12. Clinical Framework (Hyperlipidemia Specific)

Primary Condition: Hyperlipidemia / High Cholesterol **Preferred UMLS Name:** Hyperlipidemia **Semantic Type:** Disease or Syndrome **Clinical Terminologies:** MeSH: D006949 | HPO: HP:0003077 | SNOMED: 55822004 **Diagnosis Threshold:** Elevated lipid concentration in the blood requiring lipid-lowering therapy **Medical Methodology:** Standard of Care (Rosuvastatin) + Oral PCSK9 Inhibitor (Enlicitide) **Optimization Target:** Mean Percent Change in LDL-C reduction

13. Study Procedures & Methodology

Management Pathway: Standard clinical pathway for Hyperlipidemia **Education Protocol (HCP):** Site initiation visits and continuous protocol training **Education Protocol (Patient):** Informed consent process, medication administration instructions **Quality Audit Mechanism:** IP accountability logs and patient adherence tracking

14. Observation & Follow-Up Schedule

Total Duration: Approx. 24 to 52 weeks **Onsite Visit Cadence:** Screening, Baseline, Week 8, End of Study **Remote Monitoring Frequency:** Standard Q4W or Q8W as per protocol **Checklist Review Interval:** Every clinical visit

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	____	[DD-MMM-YYYY]				